

Unified constraint framework for exotic spin-dependent interactions: a matter–antimatter sector comparison

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Exotic spin-dependent interactions mediated by light, weakly coupled bosons are described in two disjoint languages: the Standard Model Extension (SME), a relativistic effective field theory of Lorentz and CPT violation, and the Dobrescu–Mocioiu (DM) classification, a complete basis of sixteen non-relativistic potentials V_1 – V_{16} in which experiments report their bounds. We construct a systematic dictionary between the two by applying the Foldy–Wouthuysen transformation to the SME fermion-sector coefficients b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$ and $H_{\mu\nu}$, and matching the resulting non-relativistic Hamiltonians onto the DM basis. These four are shown to be the complete set generating spin-dependent non-relativistic potentials: a_μ , $c_{\mu\nu}$ and e_μ appear only in spin-independent terms and f_μ not at all through third order in $1/m$, so none has an image in the DM basis. All four reductions are obtained in full and agree with the literature, including the momentum-linear d_{ij} and d_{00} components, whose apparent sign discrepancy we trace to a momentum-index convention in the reference result rather than to the derivation. We then define a dimensionless CPT asymmetry parameter $A_\alpha = (g_\alpha^f - g_\alpha^{\bar{f}})/(g_\alpha^f + g_\alpha^{\bar{f}})$ and evaluate it against a compiled database of 283 experimental constraint datasets, 247 of which enter the analysis, standardized to a common interaction-range axis. Fifteen matter–antimatter pairs can be compared, and all exhibit statistically significant asymmetry after correction for autocorrelation among interpolated points. We show analytically, and confirm across all fifteen pairs, that A_α constructed from one-sided experimental upper bounds cannot by itself distinguish CPT violation from an ordinary sensitivity gap between the two experiments. A spin-independent scalar channel in our sample shows an asymmetry comparable to the spin-dependent ones. The binding constraint on this class of CPT test is therefore the absence of antimatter-sector measurements rather than statistical power. Only 22 of 247 analyzed datasets involve an antimatter sector, and seven of twelve classified potentials have none at all. A coverage analysis identifies the e - N , n - N and p - N sectors and the V_{1a} , V_{9+10} and V_{12+13} potentials as the largest opportunities for new measurement. The mapping table, database and analysis pipeline are released as an extensible resource.

I. INTRODUCTION

The Standard Model is tested to extraordinary precision yet remains incomplete: it does not accommodate quantum gravity, offers no dark matter candidate, and does not explain the observed baryon asymmetry [1]. A well-motivated class of extensions introduces light, weakly coupled bosonic fields, such as axions, axion-like particles and dark photons, that mediate feeble long-range forces, a possibility first set out systematically by Moody and Wilczek [10]. Those that couple to fermion spin rather than to charge or mass alone are of particular interest, since spin has no classical analogue and therefore offers an unusually clean channel for probing symmetry violation.

Two formalisms describe such interactions, and they are not directly commensurable. The SME of Colladay and Kostelecký [2, 3], motivated by the spontaneous Lorentz breaking identified in string theory [4], is a relativistic effective field theory that parametrizes Lorentz and CPT violation through fixed background coefficients such as b_μ , $H_{\mu\nu}$ and $d_{\mu\nu}$; its coefficients are catalogued and constrained in the data tables of Kost-

elecký and Russell [7]. The DM classification [9] is a complete non-relativistic basis of sixteen potentials V_1 – V_{16} , generated by single-boson exchange between two fermions and recast in coordinate space for laboratory use in Ref. [21], in which essentially all laboratory searches report their limits; experiments up to 2024 are reviewed in Ref. [12]. Translating between the two requires the Foldy–Wouthuysen (FW) transformation [11], the standard technique for extracting the non-relativistic limit of a relativistic Dirac Hamiltonian.

Separately, antimatter precision physics has become one of the sharpest available probes of fundamental symmetry. Experiments at CERN compare matter and antimatter magnetic moments and spectral lines at a level that directly tests CPT invariance [14, 15]. Joining these two threads suggests a concrete observable: a CPT-violating spin-dependent interaction should appear as an asymmetry between the coupling constants inferred for matter and for antimatter.

No complete dictionary between the SME and DM bases has been published, and no framework exists for using such a dictionary to compare matter-sector and antimatter-sector constraints within a single quantitative measure. Prior mappings are partial, restricted to subsets of coefficients or potentials, and the constraint literature itself is reported in inconsistent units and conventions, which obstructs direct comparison across plat-

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forms. This paper supplies both the dictionary and the comparison framework, and reports what the currently published data can establish and what they cannot.

Section II presents the FW reductions, establishes which coefficients can contribute at all, and gives the resulting SME→DM matching table. Section III defines the asymmetry parameter and derives the central interpretive limitation on it. Section IV describes the compiled database and the statistical methodology. Sections V and VI report the matched-pair results and the coverage analysis. Section VII states the limitations of the present database, and Sec. VIII concludes.

II. SME-TO-DM MAPPING

A. The b_μ coefficient

The CPT-odd term $\mathcal{L}_b = -b_\mu \bar{\psi} \gamma_5 \gamma^\mu \psi$ separates into a spatial part b_i , which is even and therefore exact at order m^0 , and a temporal part b_0 , which is odd and enters only at order $1/m$. In the Dirac representation $\gamma^0 \gamma_5 \gamma^i = -\Sigma^i$, so projected onto the upper block

$$H_{\text{NR}}(b_i) = -\mathbf{b} \cdot \boldsymbol{\sigma} \quad (\text{matter}), \quad (1)$$

already in non-relativistic form with no further FW iteration required. The temporal component is odd, since $\gamma^0 \gamma_5 \gamma^0 = -\gamma_5$, and so contributes through the FW cross-term between the free odd operator $\boldsymbol{\alpha} \cdot \mathbf{p}$ and the perturbing odd operator $\mathcal{O}_{b_0} = -b_0 \gamma_5$. Retaining only the piece linear in b_0 ,

$$H_{\text{NR}}(b_0) = \frac{\beta}{2m} \{ \boldsymbol{\alpha} \cdot \mathbf{p}, \mathcal{O}_{b_0} \} \Big|_{\text{upper}} = -\frac{b_0(\boldsymbol{\sigma} \cdot \mathbf{p})}{m} + O(m^{-2}). \quad (2)$$

The antiparticle sector follows from the lower block by the hole-theory reinterpretation. An antiparticle of momentum $\mathbf{p}$ and spin $\mathbf{s}$ is the absence of a negative-energy solution of momentum $-\mathbf{p}$ and spin $-\mathbf{s}$, with the opposite energy. A CPT test compares a particle with its CPT-conjugate state, an antiparticle of the same momentum and reversed spin. Labelling that state by the particle's spin operator gives

$$H_{\text{NR}}^{\bar{f}}(\mathbf{p}, \boldsymbol{\sigma}) = -H_{\text{lower}}(-\mathbf{p}, \boldsymbol{\sigma}). \quad (3)$$

Since $\Sigma^i = \text{diag}(\sigma^i, \sigma^i)$ repeats the *same* operator in the upper ($\beta = +1$) and lower ($\beta = -1$) blocks,

$$\langle u | b_i \gamma^0 \gamma_5 \gamma^i | u \rangle = \langle l | b_i \gamma^0 \gamma_5 \gamma^i | l \rangle = -b_i \sigma^i, \quad (4)$$

and since b_i carries no momentum,

$$H_{\text{NR}}^{\bar{f}}(b_i) = -(-\mathbf{b} \cdot \boldsymbol{\sigma}) = +\mathbf{b} \cdot \boldsymbol{\sigma}. \quad (5)$$

The CPT-conjugate antiparticle receives the opposite shift. This matter–antimatter sign flip is the signature of a CPT-odd coefficient. It is *not* a charge-conjugation

effect on the axial-vector bilinear $\bar{\psi} \gamma_5 \gamma^\mu \psi$, which is C-even for every μ and therefore cannot by itself produce the difference between Eqs. (1) and (5). The difference is a CPT effect and appears only when CPT-conjugate states are compared.

Stated at fixed physical spin, the result looks different. An antiparticle with the particle's spin has energy $-\mathbf{b} \cdot \mathbf{s}$, the same as the particle. This is Kostecký and Lane's antiparticle rule, which leaves b_μ , $c_{\mu\nu}$ and $g_{\lambda\mu\nu}$ unchanged and reverses a_μ , $d_{\mu\nu}$, e_μ , f_μ and $H_{\mu\nu}$ [6]. The two statements are equivalent, since a spin-linear term changes sign when the spin is reversed. The CPT-conjugate comparison is the one relevant to a CPT test: with it, exact CPT symmetry gives $g_\alpha^{\bar{f}} = g_\alpha^f$. We verified both comparisons symbolically for every coefficient family, using the full FW reduction and kinetic-sector field redefinition. The CPT-conjugate comparison reproduces each coefficient's CPT parity in every family, and the fixed-spin comparison reproduces the rule of Ref. [6] in every family. Both require the momentum reversal in Eq. (3); omitting it changes the result for every term odd in $\mathbf{p}$. The result also agrees with Bluhm, Kostecký and Russell [8]. Their substitution rule from hydrogen to antihydrogen reverses a_μ , $d_{\mu\nu}$ and $H_{\mu\nu}$ but not b_μ , the fixed-spin rule. Their b_μ -dependent 1S–2S shift nevertheless changes sign, because the antihydrogen hyperfine states compared have opposite positron and antiproton spin assignments, that is, CPT-conjugate spin states.

Promoting Eq. (1) to a two-body potential by pairing the same vertex on both fermions gives the spin–spin structure

$$V_2(r) = -g_A^{(1)} g_A^{(2)} \frac{1}{4\pi} (\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) \frac{e^{-r/\lambda}}{r}, \quad (6)$$

identifying b_i with V_2 at leading order, while b_0 matches the velocity-dependent V_8 at order $1/m$.

B. The $H_{\mu\nu}$ coefficient

$H_{\mu\nu}$ is CPT-even but Lorentz-odd: CPT-conjugate matter and antimatter states receive identical shifts at tree level, so its experimental signature is a sidereal anisotropy rather than a matter–antimatter asymmetry. Its magnetic-like spatial components H_{ij} are even and generate, at order m^0 , using $\gamma^0 \sigma^{ij} \Big|_{\text{upper}} = \varepsilon^{ijk} \sigma^k$,

$$H_{\text{NR}}(H_{ij}) = +\mathcal{H}_B \cdot \boldsymbol{\sigma}, \quad \mathcal{H}_B^k = \frac{1}{2} \varepsilon^{ijk} H_{ij}, \quad (7)$$

matching the dipole–dipole potential V_3 . The electric-like mixed components H_{0i} are odd, since $\gamma^0 \sigma^{0i} = i\gamma^i$ is block-off-diagonal, and generate at order $1/m$, through the same FW cross-term used for b_0 with $\mathcal{O}_{H_E} = i \sum_k H_{0k} \gamma^k$,

$$H_{\text{NR}}(H_{0i}) = +\frac{1}{m} \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E), \quad H_E^i \equiv H_{0i}, \quad (8)$$

matching the spin–velocity potential V_7 . Applying Eq. (3) gives $H_{\text{NR}}^{\bar{f}} = H_{\text{NR}}^f$ for both components, as their

CPT-even character requires; at fixed physical spin the sign reverses, in line with the rule $H_{\mu\nu} \rightarrow -H_{\mu\nu}$ of Ref. [6]. This coefficient therefore gives $g_\alpha^f = g_\alpha^f$ and $A_\alpha = 0$ for the underlying couplings, in contrast to b_μ .

C. The $d_{\mu\nu}$ coefficient

Following Kostelecký and Lane [5], $d_{\mu\nu}$ is a CPT-even, traceless, kinetic-sector coefficient entering through the derivative coupling $\mathcal{L}_d = \frac{1}{2}i\bar{\psi}d^{\mu\nu}\gamma_5\gamma_\mu\overleftrightarrow{\partial}_\nu\psi$. With $\Gamma_\mu \equiv \gamma^0\gamma_5\gamma_\mu$ one finds $\Gamma_0 = -\gamma_5$ (odd) and $\Gamma_i = +\Sigma^i$ (even), from which

$$H_{\text{NR}}(d_{i0}) = +d_{i0} m \sigma^i \rightarrow V_2, \quad (9)$$

$$H_{\text{NR}}(d_{ij}) = +d_{ij} p^j \sigma^i \rightarrow V_7, V_8, \quad (10)$$

$$H_{\text{NR}}(d_{00}) = +d_{00}(\boldsymbol{\sigma} \cdot \mathbf{p}) \rightarrow V_8. \quad (11)$$

All three results reproduce the non-relativistic Hamiltonian of Ref. [5] exactly, including signs, once that reference's momentum convention is applied. The convention is not stated explicitly alongside their Eq. (4), which describes p_j as the three-momentum of the particle, but it is fixed unambiguously by the free-particle term of the same authors' intermediate relativistic Hamiltonian [6], $mP_0 := -p_j\gamma^0\gamma^j$. That expression equals $-\boldsymbol{\alpha} \cdot \mathbf{p}$, so recovering the free Dirac Hamiltonian $+\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m$ requires their lower-index p_j to be the covariant component $p_j = -p^j$. Read as the physical three-momentum instead, the two momentum-linear terms appear to differ by an overall sign while the mass-enhanced term still agrees. That pattern points to a momentum-index mismatch, not a derivation error. We have verified the correspondence symbolically, including an independent implementation of the kinetic-sector field redefinition $A = 1 - \frac{1}{2}\gamma^0(\Gamma_0 - \gamma_0)$ of Ref. [6].

D. The $g_{\lambda\mu\nu}$ coefficient

$g_{\lambda\mu\nu}$ is CPT-odd and dimensionless, and enters the kinetic sector through $\frac{1}{2}g_{\lambda\mu\nu}\sigma^{\lambda\mu}$ in Γ_ν . It is antisymmetric in its first two indices, since $\sigma^{\lambda\mu}$ is. Because it modifies Γ_0 , the same field redefinition used for $d_{\mu\nu}$ applies; we verified that the resulting Hamiltonian is Hermitian before reducing it.

The reduction separates by index family. Writing $\tilde{g}_j \equiv \frac{1}{2}\varepsilon_{jkl}g_{kl0}$ for the dual of the spatial-spatial, $\nu = 0$ block, and $(\mathbf{g}_{00})_k \equiv g_{k00}$,

$$H_{\text{NR}}(g_{[kl]0}) = -m \tilde{\mathbf{g}} \cdot \boldsymbol{\sigma} \rightarrow V_2, \quad (12)$$

$$H_{\text{NR}}(g_{[0k]0}) = +\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{g}_{00}) \rightarrow V_7, \quad (13)$$

$$H_{\text{NR}}(g_{[kl]j}) = +\frac{1}{2}\varepsilon_{klm}g_{mlj} p^j \sigma^k \rightarrow V_7, V_8. \quad (14)$$

The remaining family $g_{[0k]j}$ gives no contribution at this order.

Applying Eq. (3) to each family gives an exact sign flip for all three contributing families, as the CPT-odd character of $g_{\lambda\mu\nu}$ requires. The momentum reversal is essential here: the $g_{[0k]0}$ and $g_{[kl]j}$ terms are linear in $\mathbf{p}$, and comparing the lower block at the same momentum instead would wrongly suggest that these two families do not flip. At fixed physical spin, all three families give identical particle and antiparticle shifts, in line with the rule $g_{\lambda\mu\nu} \rightarrow +g_{\lambda\mu\nu}$ of Ref. [6]. The matter-antimatter signature is therefore set by the coefficient's CPT parity, for $g_{\lambda\mu\nu}$ as for b_μ .

The three surviving structures reproduce those already obtained for the other coefficients: the $\nu = 0$ spatial-spatial block is Zeeman-like and mass-enhanced, as b_i and d_{i0} are; the $\nu = 0$ temporal-spatial block is the spin-velocity form found for H_{0i} ; and the $\nu = j$ spatial-spatial block has the general $\tilde{g}_{kj}p^j\sigma^k$ shape found for d_{ij} . All three agree with Ref. [5] term by term, under the same covariant momentum convention established in Sec. II for $d_{\mu\nu}$. Since that convention was fixed without reference to g , this is an independent check on it.

E. Coefficients that generate no spin-dependent terms

The four coefficients treated above exhaust the spin-dependent sector. Splitting the non-relativistic Hamiltonian of Ref. [5] into its top-level terms, every term carrying a factor of $\boldsymbol{\sigma}$ contains only b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$ and $H_{\mu\nu}$. The coefficients a_μ , $c_{\mu\nu}$ and e_μ appear exclusively in spin-independent terms, and f_μ does not appear in the non-relativistic Hamiltonian at all through third order in $1/m$.

This is why the dictionary presented here is complete rather than partial: the Dobrescu-Mocioiu basis is spin-dependent apart from V_1 , so the excluded coefficients have no image in it at these orders. The $g_s g_s \cdot V_1 \cdot e^-e^+$ pair of Sec. V lies in this spin-independent sector, which receives contributions from a_μ , $c_{\mu\nu}$ and e_μ but from none of the four coefficients above.

F. Master matching table

Table I synthesizes the four reductions. Its final column gives the relation between the particle and antiparticle couplings for CPT-conjugate states. It describes the underlying couplings, not the published bounds: as Sec. III shows, an A_α built from one-sided bounds cannot reveal it.

III. THE CPT ASYMMETRY PARAMETER

For a coupling α constrained independently in a matter sector and its conjugate antimatter sector, we define the

TABLE I. Master SME→DM matching table, covering the complete set of SME coefficients that generate spin-dependent non-relativistic potentials (b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$, $H_{\mu\nu}$); see Sec. II E for why the remainder are excluded. Here $\tilde{g}_j \equiv \frac{1}{2}\varepsilon_{jkl}g_{kl0}$ and $(\mathbf{g}_{00})_k \equiv g_{k00}$. The final column relates particle and antiparticle couplings for CPT-conjugate states (antiparticle spin reversed), verified symbolically for every row; it follows the coefficient's CPT parity without exception. At fixed physical spin every sign in that column reverses [6]. Neither relation gives $|A_\alpha| \rightarrow 1$: a flip makes the signed A_α diverge [Eq. (16)] and equal couplings give $A_\alpha = 0$, so a value near unity in the data reflects a sensitivity gap between independent bounds (Sec. III).

SME coeff.	CPT	Non-relativistic Hamiltonian	DM potential(s)	1/ m order	Matter–antimatter
b_i	Odd	$-\mathbf{b} \cdot \boldsymbol{\sigma}$ (matter); $+\mathbf{b} \cdot \boldsymbol{\sigma}$ (CPT-conj. antimatter)	V_2	m^0	$g^f = -g^{\bar{f}}$
b_0	Odd	$-b_0(\boldsymbol{\sigma} \cdot \mathbf{p})/m$	V_8	m^{-1}	$g^f = -g^{\bar{f}}$
H_{ij}	Even	$+\mathcal{H}_B \cdot \boldsymbol{\sigma}$	V_3	m^0	$g^f = g^{\bar{f}}$
H_{0i}	Even	$+\frac{1}{m}\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E)$	V_7	m^{-1}	$g^f = g^{\bar{f}}$
d_{i0}	Even	$+d_{i0}m\sigma^i$	V_2	m^{+1}	$g^f = g^{\bar{f}}$
d_{ij}	Even	$+d_{ij}p^j\sigma^i$	V_7, V_8	m^0 (vel.)	$g^f = g^{\bar{f}}$
d_{00}	Even	$+d_{00}(\boldsymbol{\sigma} \cdot \mathbf{p})$	V_8	m^0 (mom.)	$g^f = g^{\bar{f}}$
$g_{[kl]0}$	Odd	$-m\tilde{\mathbf{g}} \cdot \boldsymbol{\sigma}$	V_2	m^{+1}	$g^f = -g^{\bar{f}}$
$g_{[0k]0}$	Odd	$+\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{g}_{00})$	V_7	m^0	$g^f = -g^{\bar{f}}$
$g_{[kl]j}$	Odd	$+\frac{1}{2}\varepsilon_{klm}g_{mlj}p^j\sigma^k$	V_7, V_8	m^0 (vel.)	$g^f = -g^{\bar{f}}$

dimensionless asymmetry

$$A_\alpha(\lambda) = \frac{g_\alpha^f(\lambda) - g_\alpha^{\bar{f}}(\lambda)}{g_\alpha^f(\lambda) + g_\alpha^{\bar{f}}(\lambda)}, \quad (15)$$

evaluated as a function of interaction range λ , where the antifermion coupling refers to the CPT-conjugate state (same momentum, reversed spin). With this definition, exact CPT invariance gives $g_\alpha^{\bar{f}} = g_\alpha^f$ and hence $A_\alpha = 0$.

It is tempting to substitute the signed relation $g_{\bar{f}} = -g_f$ implied by Eqs. (1) and (5) directly into Eq. (15). Doing so gives

$$A_\alpha = \frac{g - (-g)}{g + (-g)} = \frac{2g}{0} \rightarrow \infty, \quad (16)$$

The ratio diverges (we confirmed this with a symbolic limit); it does not approach unity.

This is not, however, the quantity that published data can supply. The inputs g_f and $g_{\bar{f}}$ are independent, strictly positive experimental *upper bounds*, not signed measurements of one underlying coupling. For any two independent positive bounds g_1, g_2 the ratio $(g_1 - g_2)/(g_1 + g_2)$ approaches ± 1 whenever one bound is far tighter than the other, *regardless of the underlying physics*: setting $g_{\text{tight}} = r g_{\text{loose}}$ and taking $r \rightarrow 0^+$ gives $A_\alpha \rightarrow 1$ with no CPT content at all.

Consequently an observed $|A_\alpha| \approx 1$ is *consistent with*, but is not *evidence for*, a non-zero b_μ : the same numerical pattern is what independent bounds of very different sensitivity produce even under exact CPT symmetry. This governs the interpretation of every asymmetry reported below, and we treat it as one of the main results of this work, not a side remark. Its practical consequence is that the most informative pairs are those with the *lowest* asymmetry, since those are the pairs in which the two sectors are probed at comparable sensitivity.

IV. DATABASE AND METHODS

A. Compilation and standardization

We compiled 283 experimental constraint datasets from the published literature, drawing on the compilation of Ref. [12] and the underlying primary sources, spanning nitrogen-vacancy-center magnetometry [27, 31, 32], torsion balances [25, 26, 28, 30, 34], atomic comagnetometers [35], precision spectroscopy of simple atoms and molecules [17, 20, 23, 33], Casimir-force measurements [18, 19], and antimatter and exotic-atom spectroscopy, including positronium and antiprotonic helium [16, 22, 24]. Each dataset is a digitized exclusion curve $g(\lambda)$ or $\alpha(\lambda)$.

Datasets are parsed for coupling family, DM potential label, fermion sector, and source. A bare numeric prefix in the source filenames denotes the potential number itself, so a 3-prefixed file bounds V_3 alone and only a 23 prefix denotes the $V_2 + V_3$ combination. Every potential assignment was validated against the source compilation's own curation records; among the 194 datasets those records cover there is no disagreement. Interaction ranges reported variously in meters, MeV^{-1} , eV^{-1} , centimeters, nanometers and femtometers are converted to a common meter scale by a unit-detection routine applied before any downstream analysis. Bounds published as a gravity-normalized Yukawa strength $|\alpha|$ are kept separate from those published as a coupling $|g|$; typical values of the two differ by more than thirty orders of magnitude.

B. Exclusions

Of the 283 compiled datasets, 36 are held out of the analysis, leaving 247. Twenty-eight are combined envelope curves, twelve in g_{AGV} and sixteen in $g_{p g_s}$, that ag-

gregate several experiments; the source compilation assigns them no single potential and flags them for review, and no single V_n applies to an envelope by construction. One is an astrophysical e - N bound with no potential assignment that can be established from the source. Two are contributor-template placeholders. Five are the 90% confidence-level variants of the Cong *et al.* hydrogen bounds [13], which the source ships alongside the 95% CL curves; they are the same measurement at a second confidence level, so retaining both would double-count the experiment. Every exclusion is recorded individually with a stated reason, and the registry retains all 283 rows so that the compiled and analyzed counts remain separately auditable.

C. Pairing and statistics

Two datasets are matched into a matter–antimatter pair only when they share coupling family, DM potential label and interaction class, their fermion sectors are physically conjugate (for example ee with $e\bar{e}$), and exactly one is flagged as antimatter. This logic is conservative: it will not match datasets differing in potential classification even where their sectors are conjugate.

For each matched pair both curves are interpolated logarithmically in $\log \lambda$ – $\log g$ onto a common log-spaced grid of 300 points spanning the overlapping λ range, and $A_\alpha(\lambda)$ is evaluated at every point. Significance is assessed in four stages, each addressing a limitation of the previous one:

1. A uniform χ^2 test assuming a flat 10% fractional uncertainty at every grid point.
2. A weighted χ^2 test estimating a per-point fractional uncertainty from the local curvature of each curve in log–log space, on the grounds that a smoothly varying published bound is better determined than one with sharp local features.
3. Both tests above treat the 300 grid points as independent, which is unjustified: they are interpolated from far fewer real measurements along smooth curves, so adjacent points are strongly correlated. An effective number of degrees of freedom is estimated from the autocorrelation length τ of the χ^2 residuals in grid-index space (the lag at which the residual autocorrelation first falls below $1/e$), as $n_{\text{eff}} = n/\tau$. The weighted χ^2 is scaled down by the same factor, $\chi_{\text{eff}}^2 = \chi^2 n_{\text{eff}}/n$, and compared with a χ^2 distribution on n_{eff} degrees of freedom. Most of the resulting p -values lie far below double precision, so we quote the equivalent one-sided Gaussian significance Z .
4. A parametric bootstrap (2000 resamples) gives a 95% confidence interval on the mean $|A_\alpha|$. Each resample perturbs both interpolated couplings by their estimated fractional uncertainties

(log-normally, so they stay positive) and recomputes A_α exactly.

Two points about the bootstrap matter for how its intervals are read. Because $|A_\alpha|$ is nonlinear in the couplings, folded at zero and saturating at one, the resampled means sit slightly off the point estimate even when the perturbations are symmetric. We remove that offset, estimated as the median bootstrap bias, before quoting the interval. And the interval only reflects the spread implied by the assumed curve uncertainties; for $|A_\alpha|$ an interval that excludes zero is not, on its own, evidence of a non-zero asymmetry.

All analysis is performed by SPINDEP, an open, version-controlled pipeline; the full analysis is reproducible from a single command, and the parsing, matching and statistical modules are covered by an automated test suite.

V. RESULTS

A. The compiled database

The 247 analyzed datasets divide by coupling family as g_{AGA} (66), $g_s g_s$ (49), $g_V g_V$ (44), $g_p g_s$ (37), g_{AGV} (25), $g_p g_p$ (15) and an unlabeled V_1 family (11). By DM potential the database is dominated by V_{4+5} (61), followed by V_1 (47), V_3 (39), V_{9+10} (31), V_2 (20) and V_{12+13} (16), with V_{1a} (14), V_{11} (7), V_8 (6), V_{2+3} (2), V_{16} (2) and V_{15} (2) making up the remainder. Several labels combine adjacent catalogue potentials that the underlying experiments do not distinguish; V_6 , V_7 and V_{14} have no classified dataset at all. Figure 1 shows the constraint atlas for V_3 , the best-covered potential in the antimatter sector.

Only 22 of the 247 analyzed datasets involve an antimatter sector: 9 in e - $\bar{p}$, 6 in e - $\bar{\mu}$ (muonium), 5 in e - e^+ , and one each in $\bar{p}$ -He and the $dd\mu^+$ molecular ion. Muonium bounds are counted on the antimatter side because the atom contains an antimuon, the same reason $dd\mu^+$ is. This imbalance, 22 antimatter datasets against 225 matter-sector ones, is the central empirical fact of this work.

B. Matched pairs

The pairing logic returns fifteen matter–antimatter pairs, listed in Table II with the mean absolute asymmetry, the autocorrelation-corrected effective degrees of freedom, the resulting significance, and the 95% bootstrap interval.

Effective degrees of freedom range from 6 to 21 across the fifteen pairs, a factor of 14–50 below the 300 grid points, so the correction is far from cosmetic. Every pair stays significant after it. The weakest is the

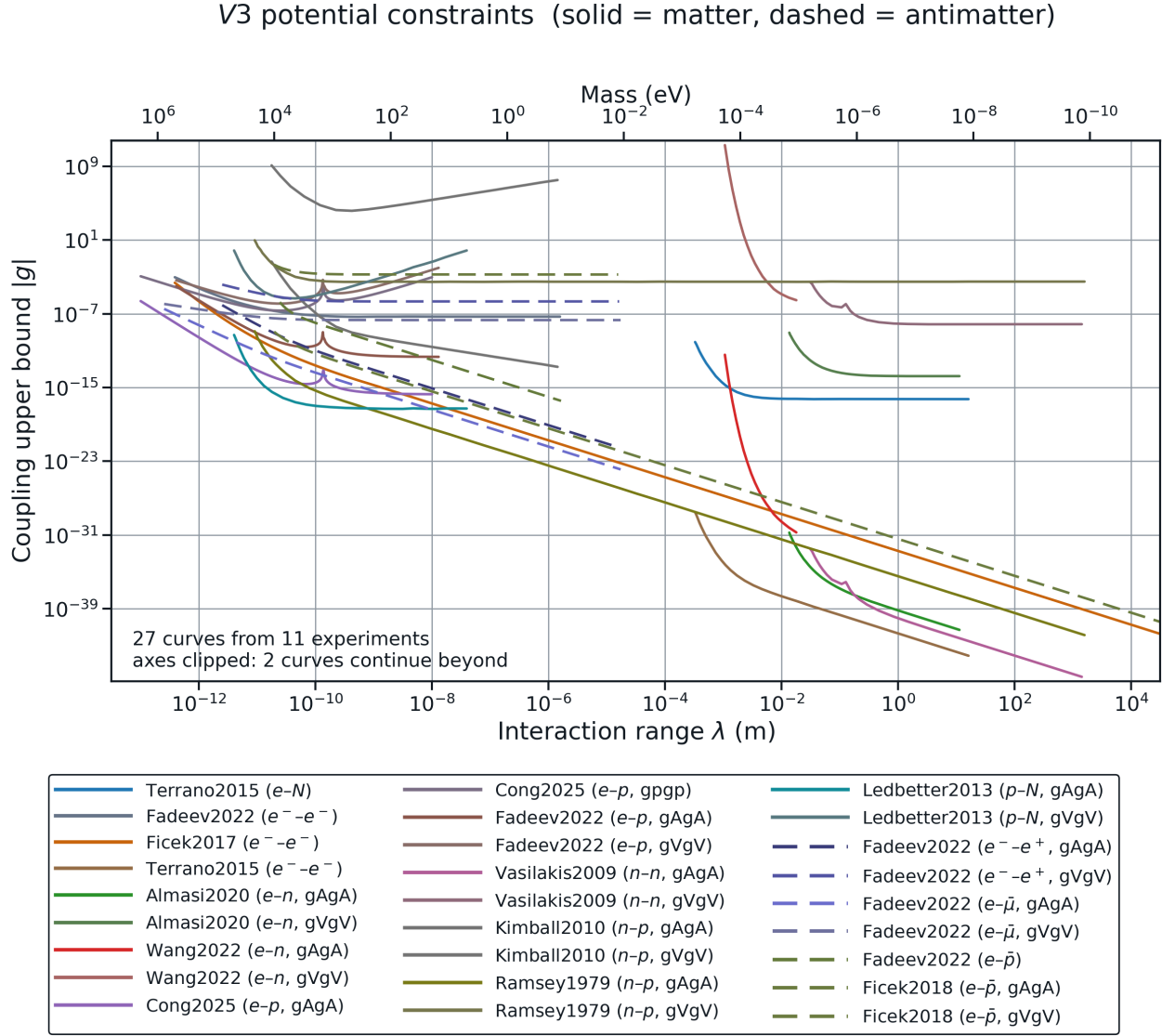


FIG. 1. Constraint atlas for V_3 , the potential with the broadest antimatter-sector coverage in the compiled database. Solid curves are matter-sector bounds, dashed curves antimatter-sector; the region above each curve is excluded. Shown at full page width because the panel carries 27 curves from 11 experiments.

Jiao/Karshenboim $e-e^+$ pair at $Z = 5.1$ ($p \approx 2 \times 10^{-7}$); the others lie between $Z = 26$ and $Z = 51$.

Two rows repeat two others. The source compilation files the same Fadeev *et al.* V_3 curves under both $g_V g_V$ and $g_p g_p$, so the $g_V g_V$ and $g_p g_p$ pairs built from them give identical numbers. The fifteen pairs therefore contain thirteen independent comparisons.

C. Interpretation

By Sec. III, an asymmetry near unity in Table II is consistent with, but not evidence for, CPT violation: a large sensitivity gap between the two measurements produces the same signature regardless of the underlying physics. This applies to every row, including the three $e-\bar{p}$ pairs

at 0.9998 and above at the top of the table.

The $g_s g_s \cdot V_1 \cdot e-e^+$ pair at 0.8727 makes the same point from a different direction. V_1 is spin-independent, so none of the coefficients in Table I feeds it, yet its asymmetry sits well inside the range spanned by the g_{AgA} pairs (0.33 to 1.00). A high asymmetry is simply what two bounds of very different sensitivity produce, whatever channel they sit in. Figures 2 and 3 contrast the highest- and lowest-asymmetry pairs.

The lowest entry, 0.3336 for the $g_{AgA} \cdot V_2 \cdot e-e^+$ pair of Refs. [27, 29], is the pair whose two bounds are closest in sensitivity, and is therefore the single most informative datum currently available for constraining b_μ in the $e-e^+$ channel, because it is not saturated.

The four pairs built on the recent hydrogen-spectroscopy bounds of Ref. [13] show the effect directly.

TABLE II. All fifteen matched matter–antimatter pairs, with the autocorrelation-corrected effective degrees of freedom dof_{eff} , the weighted- χ^2 significance Z after that correction, and the 95% bootstrap confidence interval on the mean asymmetry. Per Sec. III, a value near unity is consistent with, but not evidence for, CPT violation.

Coupling	Potential	Sector	Matter / antimatter source	$ \overline{A}_\alpha $	dof_{eff}	Z	95% CI
g_{AgA}	V_3	$e\text{-}\bar{p}$	Cong [13] / Fadeev [22]	1.0000	8	51.1	[1.0000, 1.0000]
g_{AgA}	V_2	$e\text{-}\bar{p}$	Cong [13] / Ficek [24]	0.9999	21	34.4	[0.9999, 0.9999]
g_{AgA}	V_2	$e\text{-}\bar{p}$	Karshenboim [29] / Ficek [24]	0.9998	17	38.8	[0.9998, 0.9998]
g_{AgA}	V_2	$e\text{-}e^+$	Ficek [23] / Karshenboim [29]	0.9892	21	48.2	[0.9886, 0.9896]
g_{AgA}	V_3	$e\text{-}e^+$	Ficek [23] / Fadeev [22]	0.9539	6	38.5	[0.9518, 0.9557]
$g_V g_V$	V_3	$e\text{-}e^+$	Fadeev [22] / Fadeev [22]	0.9535	6	37.1	[0.9516, 0.9552]
$g_p g_p$	V_3	$e\text{-}e^+$	Fadeev [22] / Fadeev [22]	0.9535	6	37.1	[0.9516, 0.9552]
$g_p g_p$	V_3	$e\text{-}\bar{p}$	Cong [13] / Ficek [24]	0.9304	6	34.3	[0.9293, 0.9314]
g_{AgA}	V_3	$e\text{-}\bar{p}$	Cong [13] / Ficek [24]	0.9303	6	34.3	[0.9292, 0.9313]
$g_s g_s$	V_1	$e\text{-}e^+$	Delaunay [20] / Adkins [16]	0.8727	21	25.7	[0.8686, 0.8770]
g_{AgA}	V_3	$e\text{-}\bar{p}$	Fadeev [22] / Ficek [24]	0.8236	6	37.3	[0.8168, 0.8299]
g_{AgA}	V_3	$e\text{-}\bar{p}$	Fadeev [22] / Fadeev [22]	0.8044	8	43.8	[0.8024, 0.8063]
$g_V g_V$	V_3	$e\text{-}\bar{p}$	Fadeev [22] / Ficek [24]	0.7994	7	31.8	[0.7971, 0.8015]
$g_p g_p$	V_3	$e\text{-}\bar{p}$	Fadeev [22] / Ficek [24]	0.7994	7	31.8	[0.7971, 0.8015]
g_{AgA}	V_2	$e\text{-}e^+$	Jiao [27] / Karshenboim [29]	0.3336	9	5.1	[0.3197, 0.3474]

Each can be set against a pair with the same coupling, potential and antimatter partner but an older $e\text{-}p$ bound on the matter side, and in all four the newer bound gives the higher asymmetry: 0.9999 against 0.9998 for $g_{AgA} \cdot V_2$ with Ficek *et al.*, 1.0000 against 0.8044 for $g_{AgA} \cdot V_3$ with Fadeev *et al.*, and 0.9303 against 0.8236 (g_{AgA}) and 0.9304 against 0.7994 ($g_p g_p$) for V_3 with Ficek *et al.* Improving the matter-sector measurement, with no change to the antimatter sector and none to the underlying physics, drove the asymmetry *up*. This is a direct empirical confirmation of the argument of Sec. III: A_α built from one-sided bounds tracks the sensitivity ratio, not CPT status.

VI. COVERAGE GAP ANALYSIS

A. By fermion sector

Table III compares, for each of the eleven canonical matter–antimatter *sector* pairs, the number of analyzed datasets on each side. These are pairings of fermion sectors, not the matched dataset pairs of Sec. V.

The three nucleus sectors dominate the gap. $e\text{-}N$ is the single largest opportunity, with 50 matter-sector datasets against zero in $e\text{-}\bar{N}$, but $n\text{-}N$ is barely behind at 49 and $p\text{-}N$ adds a further 18; between them these three account for 117 of the 225 matter-sector datasets and carry no antimatter-sector counterpart at all. The $e\text{-}n$, $n\text{-}p$ and $n\text{-}n$ sectors each have a smaller but still substantial base (8–16 datasets) with no counterpart. The $e\text{-}\mu$ row is the mirror image: all six of its bounds come from muonium and so sit on the antimatter side, with no matter-sector $e\text{-}\mu$ bound to compare them with.

The eleven sector pairs account for 208 matter-sector and 20 antimatter-sector datasets, 228 in all, of the 247 analyzed. The remaining 19 do not belong to a matter–

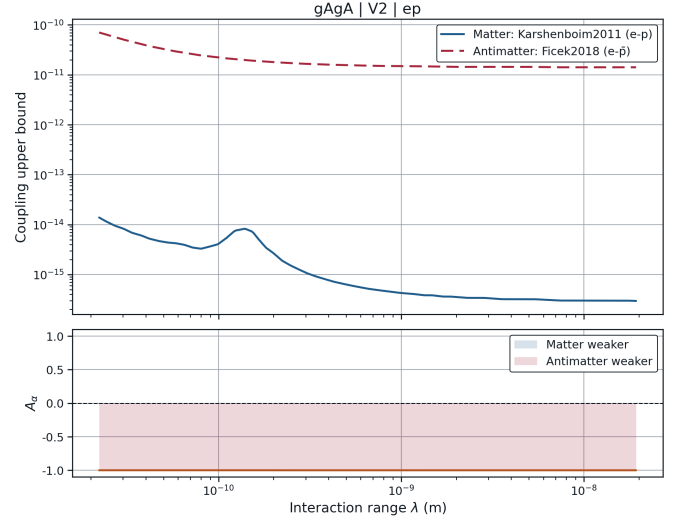


FIG. 2. Matter–antimatter comparison for the $g_{AgA} \cdot V_2 \cdot e\text{-}\bar{p}$ pair of Refs. [24, 29], one of the three highest asymmetries in the database ($|\overline{A}_\alpha| = 0.9998$). The two bounds differ by orders of magnitude in sensitivity over the overlapping range, which is sufficient on its own to saturate A_α .

antimatter sector pair: 16 are astrophysical bounds quoted for a single sector with no laboratory counterpart, and three are the $\mu\text{-}N$, $\bar{p}\text{-He}$ and $dd\mu^+$ entries, whose sectors have no counterpart row. The last two of these are also why the antimatter column totals 20 rather than 22.

B. By potential

Table IV breaks the same comparison down by DM potential. Seven of the twelve classified potentials (V_{11} , V_{15} , V_{16} , V_{1a} , V_{9+10} , V_{12+13} and V_{2+3}) have *zero* antimatter–

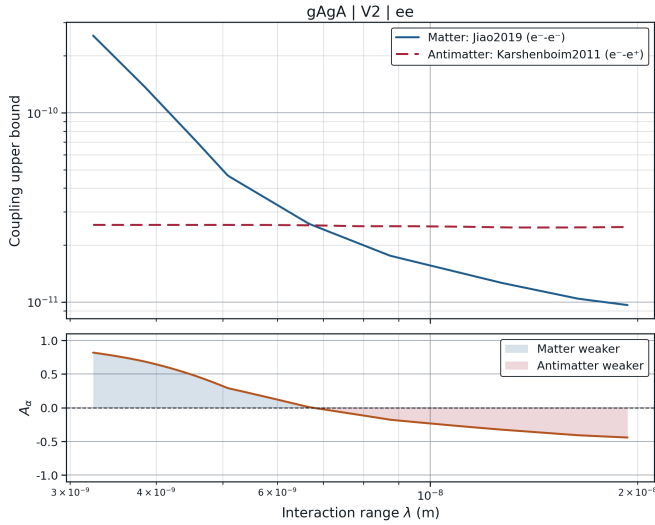


FIG. 3. Matter–antimatter comparison for the $g_{AgA} \cdot V_2 \cdot e-e^+$ pair, the lowest asymmetry in the database ($|A_\alpha| = 0.3336$). The two sectors are probed at comparable sensitivity here, making this the most informative pair in the present database despite its smaller asymmetry.

TABLE III. Matter versus antimatter dataset counts per canonical fermion sector pair. ^aA $\bar{p}$ - N bound does exist, from antiprotonic-helium spectroscopy [33], but it is published as a gravity-normalized strength $|\alpha|$ rather than a coupling $|g|$ and cannot be compared with the p - N datasets without a conversion not performed here. The gap in comparable data is real, but it comes from the units, not from an absence of measurement. ^bAll current e - μ bounds come from muonium, which contains an antimuon.

Matter	Antimatter	Matter	Anti	Status
$e-e$	$e-e^+$	40	5	Populated
$e-p$	$e-\bar{p}$	17	9	Populated
$e-\mu$	$e-\bar{\mu}$	0	6	Antimatter only ^b
$e-n$	$e-\bar{n}$	16	0	Gap
$\mu-\mu$	$\mu-\bar{\mu}$	1	0	Gap (thin both sides)
$n-p$	$n-\bar{p}$	9	0	Gap
$n-n$	$n-\bar{n}$	8	0	Gap
$p-p$	$p-\bar{p}$	0	0	No data either side
$e-N$	$e-\bar{N}$	50	0	Largest gap
$n-N$	$n-\bar{N}$	49	0	Gap (second largest)
$p-N$	$p-\bar{N}$	18	0 ^a	Gap in comparable units

sector datasets. Only V_3 has antimatter coverage exceeding a handful, which is why fourteen of the fifteen matched pairs involve V_2 or V_3 .

Figures 4 and 5 display the same structure graphically. Both count *independent bounds* rather than dataset files: 48 of the 247 analyzed datasets are the same measured bound filed under more than one coupling family, and are collapsed to a single curve before plotting, leaving 199 independent bounds. Totals read off these figures are therefore systematically smaller than the counts in Tables III and IV, which count files. The two views agree on ev-

TABLE IV. Matter versus antimatter dataset counts per Dobrescu–Mocioiu potential.

Potential	Matter	Anti	Status
V_{4+5}	58	3	Thin antimatter coverage
V_{9+10}	31	0	Gap
V_3	29	10	Best covered
V_{1a}	14	0	Gap
V_2	17	3	Thin antimatter coverage
V_1	42	5	Thin antimatter coverage
V_{12+13}	16	0	Gap
V_{11}	7	0	Gap
V_8	5	1	Thin antimatter coverage
V_{15}	2	0	Gap
V_{16}	2	0	Gap
V_{2+3}	2	0	Gap (combination label)

ery gap; they differ only in how multiply-filed bounds are counted.

C. Proposed experimental priorities

Three directions follow. First, antimatter-sector effort is best prioritized in the e - N , n - N , p - N , e - n , n - p and n - n sectors, which together account for 150 of the 225 matter-sector datasets and zero antimatter-sector ones; precision antiproton experiments such as BASE [14] and antihydrogen spectroscopy such as ALPHA [15] are the nearest existing platforms. Second, the same logic applies at the potential level to V_{1a} , V_{9+10} and V_{12+13} , each with 14–31 matter-sector datasets and none on the antimatter side; V_1 , V_2 and V_8 have thin but non-zero coverage and are better placed as candidates for improving existing precision than for opening a channel from zero. Third, following the precedent of Ref. [36], reviewed in Ref. [12], in which co-magnetometer data taken for sidereal Lorentz- and CPT-violation searches were re-analyzed as new bounds on the V_{12+13} , V_{4+5} and V_{9+10} terms, a systematic reanalysis of existing BASE, ALPHA and co-magnetometer data is likely to close part of the gap faster, and at lower cost, than commissioning new measurements.

Across all three, the asymmetry argument of Sec. III implies a specific design criterion: a new antimatter-sector bound is scientifically useful for a CPT test in proportion to how closely it *matches* the sensitivity of the corresponding matter-sector bound, not how tight it is in absolute terms. A bound many decades looser than its matter-sector counterpart will saturate A_α and carry no CPT information.

VII. LIMITATIONS

Two limitations of the present database are distinct from the physics coverage gaps above.

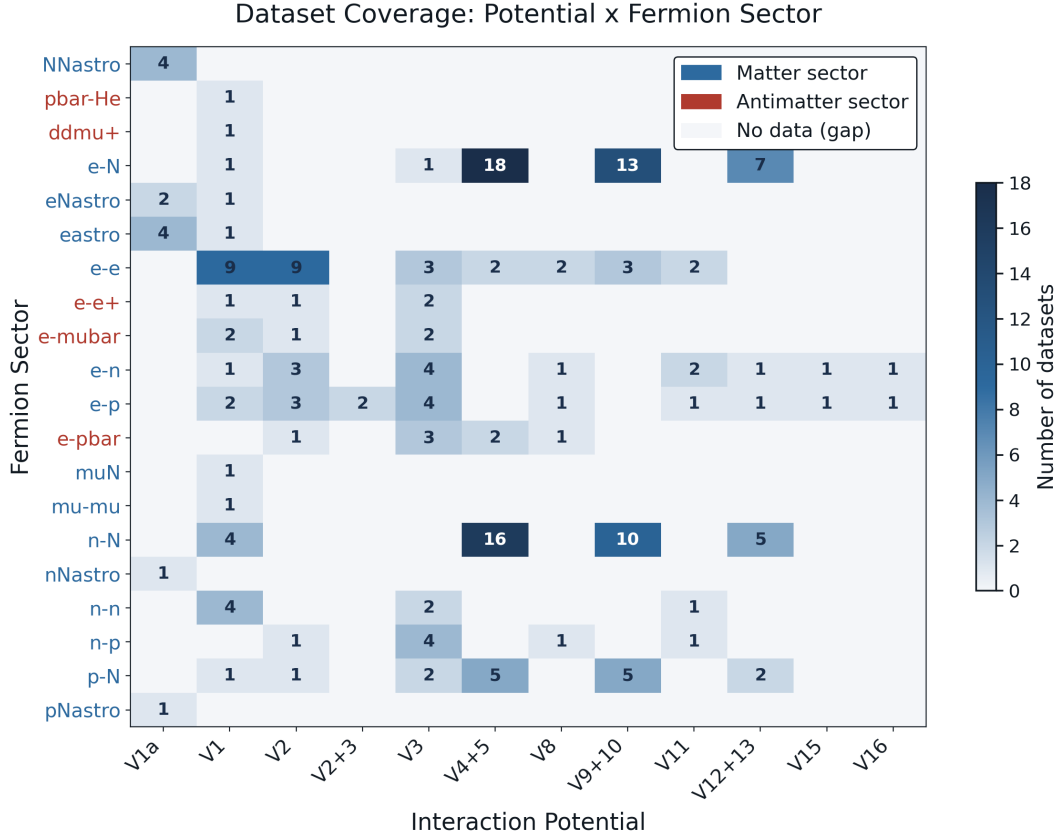


FIG. 4. Dataset coverage by potential and fermion sector; antimatter-sector rows are highlighted. Cells count independent bounds, not dataset files.

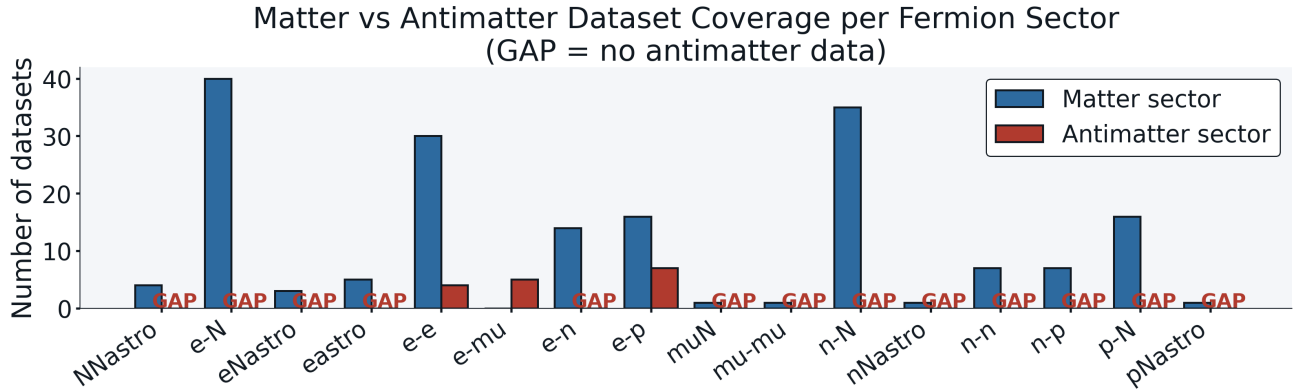


FIG. 5. Matter versus antimatter dataset coverage per fermion sector. Sectors marked GAP have no antimatter-sector data at all.

First, the 36 datasets held out of the analysis are excluded rather than reclassified, and the reasons differ in kind. Twenty-eight are combined envelope curves, twelve in g_{AgV} and sixteen in $g_p g_s$, that the source compilation marks for review and assigns no potential; these are not a parsing gap that a naming convention could close, since no single V_n describes an envelope over several experiments, and they would have to be decomposed into constituent bounds before entering a per-potential analysis.

Five are the 90% CL duplicates of the Cong *et al.* hydrogen bounds described in Sec. IV. Two are contributor-template placeholders that were previously being read as V_1 and V_2 data on the strength of a trailing digit in their filenames. The remaining one is a $g_p g_p$ astrophysical $e-N$ bound. It is a real measurement, but its potential is unassigned upstream, and assigning V_{1a} from the filenames of related datasets would be a guess, not evidence.

Second, six entries under V_3 , V_{9+10} and V_{15} report in-

interaction ranges that are physically implausible, reaching $\lambda \sim 10^{52}$ m against an observable universe of order 10^{26} m, almost certainly from a mis-detected unit token. Three of the fifteen matched pairs draw on an affected dataset: the $g_{AGA} \cdot V_3 \cdot e-e^+$ pair on the matter side, and the two $g_{AGA} \cdot V_3 \cdot e-\bar{p}$ pairs that use the bound of Ref. [24] on the antimatter side. None of these results is affected, because the asymmetry is evaluated only on the overlapping range of the two curves and in every case the implausible tail lies outside it: the ranges used are $[2.5 \times 10^{-12}, 1.6 \times 10^{-5}]$ m, $[2.0 \times 10^{-11}, 1.0 \times 10^{-8}]$ m and $[2.0 \times 10^{-11}, 1.3 \times 10^{-8}]$ m. A broader coverage claim drawn from the full database without this cleanup would nonetheless not be reliable.

VIII. CONCLUSIONS

We have constructed a systematic dictionary between the SME fermion-sector coefficients b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$ and $H_{\mu\nu}$ and the Dobrescu–Mocioiu potential basis by explicit Foldy–Wouthuysen reduction, summarized in Table I. These four are the complete set generating spin-dependent non-relativistic potentials, so the dictionary is closed rather than partial: a_μ , $c_{\mu\nu}$ and e_μ contribute only spin-independent terms and f_μ none at all through third order in $1/m$. All four reductions are complete and agree with established results. The apparent sign discrepancy in the momentum-linear d_{ij} and d_{00} components against Ref. [5] is traced to the momentum-index convention of that reference, fixed by the free-particle term of the corresponding derivation [6], and is not a feature of the reduction.

Using this dictionary we defined a CPT asymmetry parameter and evaluated it against the 247 analyzed constraint datasets. Fifteen matter–antimatter pairs could

be compared, and all show statistically significant asymmetry after autocorrelation correction. We showed, however, that this significance cannot be read as evidence of CPT violation: an asymmetry built from one-sided upper bounds saturates whenever the two experiments differ in sensitivity, and a spin-independent scalar channel in our sample shows an asymmetry comparable to the spin-dependent ones. The practical consequence is a design criterion: what makes an antimatter-sector bound useful for a CPT test is how well its sensitivity matches the matter-sector bound, not how tight it is in absolute terms.

The more important result is what this shows about the data. Only 22 of 247 analyzed datasets involve an antimatter sector; seven of twelve classified potentials and six of eleven fermion-sector pairs have zero antimatter-sector coverage despite substantial matter-sector data. The binding constraint on testing CPT symmetry through exotic spin-dependent interactions is therefore not matter-sector precision, nor statistical power, but the near-total absence of comparably precise antimatter-sector measurements in the very channels where matter-sector data is most abundant. We offer the mapping table, the compiled database and the analysis pipeline as an extensible resource for closing that gap, designed so that future antimatter-sector measurements are incorporated as they are published.

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